MILITARY
SPECIFICATION

MIL-A-55101A(SigC)
31 August 1962
Superseding MIL-A-55101(SigC)
16 May 1960

ANTENNA AT-912( )/VRC

1. SCOPE

1.1 This specification covers one type of vehicular-mounted whip antenna for use with a frequency modulation (fm) vehicular communications system operating over the frequency range of 30.00 to 75.95 mc. The antenna is designated as Antenna AT-912( )/VRC and consists of the following units (see 6.7):

Antenna Matching Unit MX-2799( )/VRC
Antenna Element AT-1095( )/VRC
Antenna Element AT-1096( )/VRC
Base, Antenna Support AB-719( )/VRC

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-V-173	Varnish, Moisture-and-Fungus-Resistant, for the Treatment of Communications, Electronic, and Associated Electrical Equipment.
MIL-T-713	Twine and Tape, Lacing and Tying (For Use in Electrical and Electronic Equipment).
MIL-S-901	Shockproof Equipment, Class HI (High Impact), Shipboard.
MIL-I-6867	Impregnants for Aluminum and Magnesium.

FSC 5820

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SPECIFICATIONS (Cont'd)

MILITARY

MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-169	Extreme-Temperature Cycle.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-276	Impregnation of Porous Nonferrous Metal Castings.

DRAWINGS

SIGNAL CORPS

SC-A-46439	List of Accessories for Package Tester.
SC-A-46694	Materials and Processes for Printed Wiring Boards and Assemblies of the Radio Set AN/VRC-12( ) Series of Equipments.
SC-GL-57754	Drawing and Gage List for Mast Base AB-15( )/GR
SC-GL-323065	Drawing and Gage List for Antenna Element AT-1095()/VRC
SC-GL-323066	Drawing and Gage List for Antenna Matching Unit MX-2799( )/VRC.
SC-DL-415340	Drawing and Data List for Base, Antenna Support AB-719( )/VRC.
SC-DL-415380	Drawing and Data List for Antenna Element AT-1096( )/VRC.
SC-DL-415405	Drawing and Data List for Antenna Element AT-1095( )/VRC.
SC-DL-415415	Drawing and Data List for Antenna Matching Unit MX-2799( )/VRC.
SC-A-362100	Preparation of Preproduction Packs.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

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3. REQUIREMENTS

3.1 Description.- Antenna AT-912( )/VRC consists of a center-fed whip assembly and Antenna Matching Unit MX-2799( )/VRC. The whip assembly consists of Antenna Element AT-1095( )/VRC (upper section), Antenna Element AT-1096( )/VRC (lower section), and Base, Antenna Support AB-719( )/VRC. When installed in military vehicles and used with Receiver-Transmitters, Radio RT-246( )/VRC and RT-524( )/VRC, the antenna matching networks are automatically adjusted in ten bands to tune the antenna to any frequency to which the radio set is tuned. When used with Receiver-Transmitter, Radio RT-505( )/PRC-25, the tuning is accomplished by a manual bandswitch located on the panel of Amplifier-Power Supply Group OA-3633( )/VRC. The radiation current distribution is controlled by networks in the matching unit for efficient radiation for ground-to-ground communication, and to effect isolation of the antenna from vehicle. This control is achieved by means of a cable choke assembly consisting of a series inductor(herein referred to as a cable choke) between the outer conductor of the coaxial line and ground at the base of the antenna, and a tapped inductor in parallel to provide proper reactance in each band.

3.2 Preproduction samples.- The contractor shall furnish two preproduction samples of Antenna AT-912( )/VRC, or each unit thereof on order, for approval, provided the invitation for bids require samples.

3.3 Construction.- The antenna shall be constructed in accordance with the applicable drawing and data list indicated below:

Unit	Drawing and Data List
Antenna Matching Unit MX-2799( )/VRC	SC-DL-415415
Antenna Element AT-1095( )/VRC	SC-DL-415405
Antenna Element AT-1096( )/VRC	SC-DL-415380
Base, Antenna Support AB-719( )/VRC	SC-DL-415340

3.4 Printed wiring assemblies.- Materials and processes for printed wiring assemblies shall conform to Drawing SC-A-46694 unless otherwise specified in the equipment drawings.

3.5 Impregnation of aluminum die castings.- All aluminum sand and die castings shall be impregnated in accordance with Standard MIL-STD-276, using impregnating material Type II per Specification MIL-I-6869. Castings which have been impregnated shall be marked "IMP". The impregnation shall be to correct general seepage only and shall not be used to correct poor foundry techniques or significant porosity. (See 4.4.)

3.6 Mounting hardware.- Antenna Matching Unit MX-2799( )/VRC and Base, Antenna Support AB-719( )/VRC shall be furnished with the bag of mounting hardware specified in the applicable equipment drawings.

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3.7 Cleaning.-

3.7.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.7.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.8 Finish.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4.) The final paint film on Type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.3.)

3.9 Marking.-

3.9.1 General.- Marking shall conform to Specification MIL-M-13231. (See 4.4.)

3.9.2 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimun disassembly of the equipment.

3.9.3 Serial numbers.- The Antenna Matching Unit MX-2799( )/VRC shall have serial numbers.

3.10 Riveting.- After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.11 Securing of parts.-

3.11.1 General.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.11.2 Securing of parts by threaded fasteners.- Nylok or other screws with plastic devices, and Loctite or similar sealants shall not be subjected to temperatures in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.12 Soldering.-

3.12.1 Solder.- Solder shall conform to the equipment drawings.

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3.12.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for thinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.12.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall have become loosened.

3.13 Wiring and cabling.- Wiring and cabling shall be neat and sturdy.

3.13.1 Cabling.- Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by tape or twine conforming to Specification MIL-T-713. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.13.2 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes.

(a) To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnecting other parts.

(c) To provide for at least two replacements of the part to which the wire or cable is connected, except for certain radio-frequency leads which must be as short as possible for electrical reasons.

(d) To facilitate field repair of broken or cut wires.

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3.13.3 Protection.- Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.13.4 Support.- Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (i) during and after subjection of the equipment to specified service conditions or (ii) after service or repair of the equipment in a normal manner.

3.13.5 Clearance.- Clearance between wires or cables and parts shall be provided to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.13.6 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.13.7 Connections.-

3.13.7.1 General.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and nonmetallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.14 Service conditions.- The equipment shall meet the following service conditions; where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.14.1 Operation.- Continuous use for a period of one year, under the worldwide environmental conditions specified in 3.14.2 through 3.14.8, with a duty cycle of 23 hours on and one hour off, with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc, he shall be guided by this requirement. Approval of the preproduction sampel (see 3.2) shall be considered as compliance with this requirement.

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3.14.2 Temperature (see 4.7.1).-

(a) Operating: Ambient temperature in the range of +150°F to -40°F. (The 150°F temperature includes affect of sun-load.) Exposure of the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.14.3 Relative humidity(see 4.7.2).- Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.14.4 Elevation (see 4.7.3).-

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

3.14.5 Immersion (see 4.7.4).- Three feet of water for 2 hours.

3.14.6 Orientation.-

(a) Operating (see 4.7.5).- Any orientation up to 20 degrees from normal operating position (that is forward, backward, left, or right).

(b) Nonoperating: Storage in any position for a period of two years. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc, he shall be guided by this requirement. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

3.14.7 Sand and dust (see 4.7.6).- As encountered during heavy traffic over dry clay roads or in desert areas.

3.14.8 Vibration, bounce, and shock.- As described in 3.15.

3.15. Vibration, bounce, and shock.-

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3.15.1 Vibration.-

3.15.1.1 Vibration; internal, antenna matching unit (see 4.7.7.1).- When vibrated in accordance with 4.4.7.1, the amplitude of vibration of any part, subassembly, or structural member of Antenna Matching Unit MX-2799( )/VRC shall not exceed twice the amplitude of the vibration applied to the matching unit at any frequency between 10 and 55 cycles per second.

3.15.1.2 Vibration, whip assembly (see 4.7.7.2).- The whip assembly (AT-1095( )/VRC, AT-1096( )/VRC, and AB-719( )/VRC assembled) shall be vibrated at the natural frequency which produces the greatest amplitude for a period of two hours without electrical or mechanical failure.

3.15.2 Bounce and shock.- The equipment shall be capable of meeting the requirements of Table I.

Table I Bounce and shock

Paragraph heading	Paragraph No.	Performance after test
Bounce;	4.7.8	Specified performance. (Note A). No physical damage.
Shock; ballistic	4.7.9	Operable. (Note A). Any physical damage shall be minor only.
Shock; bench-handling	4.7.10	Specified performance. (Note A). No physical damage.

Note A: The equipment shall meet specified performance for the following requirements: Cable choke assembly (3.16.1), Voltage standing wave ratio (3.16.2), Feed System efficiency (3.16.3), and Power handling (3.16.6).

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3.16 Electrical requirements.-

3.16.1 Cable choke assembly (see 4.9.1).- The admittance to ground at the antenna connector of Antenna Matching Unit MX-2799( )/VRC (whip assembly disconnected) shall conform to values shown in the chart below:

Band	Conductance (Millimhos) Maximum	Susceptance $\frac{}{2\pi f}$ (uuf)	
		Max.	Min.
1	.15	9	1
2	.15	12	3
3	.15	13	2
4	.25	13	3
5	.25	9	0
6	.25	10	3
7	.25	12	5
8	.25	13	4
9	.50	17	7
10	.40	12	2

3.16.2 Voltage standing wave ratio (VSWR).(See 4.9.2).- The VSWR of the Antenna AT-912( )/VRC shall not exceed 3:1 over the operating frequency range, when measured in accordance with 4.9.2.

3.16.3 Feed system efficiency(see 4.9.3).- The rf voltage at the base of the radiator with 70 watts of rf input power applied at the rf input terminals of the antenna matching unit, shall conform to the values listed below:

Band	Minimum Voltage (rms)
1	225
2	200
3	150
4	160
5	165
6	150
7	125
8	115
9	110
10	40

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3.16.4 Input voltage range(see 4.9.4).- The Antenna AT-912( )/VRC shall be capable of meeting full specification requirements when operating from a 22.0 to 30.0 vdc source.

3.16.5 Radiation (see 4.9.5).- For equal input power, the ratio of the radiation field intensity on the ground of the antenna under test (P) relative to that of a series resonant base fed whip antenna ((P$_o$) shall conform to the values listed below.

Band	Frequency(mc)	P/P$_o$ in DB (Minimum)
1	30.1	-5
5	49.8	-2
10	73.3	0

3.16.6 Power handling (see 4.9.6).- The Antenna AT-912( )/VRC shall continuously handle input rf power of 70 watts without damage, for twenty minutes, at any frequency in its operating frequency range. When measured in accordance with 4.9.6, the reflected power shall not exceed 25 percent of the forward power.

3.17 Impact(see 4.10).- The antenna shall be capable of meeting the performance requirements specified herein, after subjection to the test of 4.10. Physical damage, if any, shall be minor only.

3.18 Flexibility(see 4.11).- The antenna shall withstand the flexing tests of 4.11 without electrical or mechanical damage.

3.19 Bounce preconditioning.- The antenna matching unit shall be capable of meeting the requirements specified herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6. (Also see 4.5.)

3.20 Air-seal.- When the Antenna matching unit is tested as specified in 4.12, the decrease in vacuum shall not exceed 0.01 pounds per square inch, guage (psig), per minute.

3.21 Interchangeability.- Like units, assemblies, subassemblies, printed wiring assemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.14). Individual items shall not be handpicked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.22 Technical literature and running spare parts.- Technical literature and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2.)

3.23 Workmanship.- The equipment shall be manufactured and assembled in accordance with the equipment drawings and the applicable portions of the following paragraphs herein:

3.7 Cleaning
3.10 Riveting
3.11 Securing of parts
3.12 Soldering
3.13 Wiring and cabling

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)

(b) Acceptance inspection..

(1) Acceptance inspection of equipment before preparation for delivery. (See 4.4 and 4.5.)

(2) Acceptance inspection of preparation for delivery. (See 4.15.)

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table II, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, and group C (see Tables III, IV, and V, respectively). The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, ballistic, and (4) immersion; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table II - Preproduction Inspection

Inspection (for additional preproduction inspection see 4.3)	Rqt Para.	Insp Para.
Moisture resistance	3.14.3	4.7.2
Elevation	3.14.4	4.7.3
Orientation	3.14.6	4.7.5
Sand and dust	3.14.7	4.7.6
Shock, bench-handling	3.15.2	4.7.10
Impact	3.17	4.10
Flexing test	3.18	4.11

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4.4 Acceptance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Printed wiring assemblies	3.4
Impregnation of aluminum castings	3.5
Finish	3.8
Marking	3.9

4.5 Acceptance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government-- at its dicretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.8.) Test equipment for Government verification inspection shall be made available by the contractor. Each Antenna Matching Unit which will be subjected to group A, group B, or group C inspection shall be bounce preconditioned after final assembly. (See 3.19).

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table III and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the air-seal test (4.12) shall be last. (See 6.6.)

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Table III - Group A inspection

Inspection	Rqt par.	Insp par.	AQL Major	AQL Minor
Visual and mechanical:	3.28	4.13	----	----
Antenna Element AT-1095()/VRC			1.0%	4.0%
Antenna Element AT-1096()/VRC			1.0%	4.0%
Base, Antenna Support AB-719( )/VRC:			1.0%	4.0%
Antenna Matching Unit MX-2799( )/VRC:			----	----
Case assembly			1.0%	4.0%
Impedance matching network assemblies (plug-in boards)			0.65%	2.5.%
Matching Unit assembly (less case)			1,5 dphu	6.5 dphu
Assembled MX-2799( )/VRC			1.0%	4.0%
Electrical:				
Cable choke assembly	3.16.1	4.9.1	1.5 dphu for subgroup	
Voltage standing wave ratio	3.16.2	4.9.2		
Air Seal	3.20	4.12	1.0%	

dphu - Defects per hundred units

4.5.2 Group B inspection.- This inspection, including sampling shall conform to Table IV and to the special procedures for small sample inspection of Standard MIL-STD-105. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection.

4.5.2.1 Group B sampling plans.- The group B sampling plans, for the AQL's listed in Table IV, shall be as follows:

AQL	Inspection level for normal and tightened inspection	Inspection level for reduced inspection
2.5%	L-8	L-6
4.0%	L-8	L-6
6.5%	L-7	L-5

4.5.2.2 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

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Table IV - Group B inspection

Inspection	Rqt par.	Insp par.	AQL (See 4.5.2.1)
Feed system efficiency	3.16.3	4.9.3	4.0%
Input voltage range	3.16.4	4.9.4	4.0%
Interchangeability (dimensions)	3.21	4.14(a)	6.5%
Interchangeability(plug-in impedance matching network assembly performance)	3.21	4.14(b)	2.5%

4.5.3 Group C inspection.- This inspection shall be as listed in Table V, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table V - Group C inspection

Inspection	Rqt par.	Insp par.
SUBGROUP I (see 4.5.3.1.1)		
Immersion	3.14.5	4.7.4
SUBGROUP II (see 4.5.3.1.2)		
Radiation	3.16.5	4.9.5
Power handling	3.16.6	4.9.6
SUBGROUP III (see 4.5.3.1.3)		
Temperature	3.14.2	4.7.1
Vibration:		
Vibration; internal, matching unit	3.15.1.1	4.7.7.1
Vibration; whip assembly	3.15.1.2	4.7.7.2
Bounce	3.15.2	4.7.8
Shock, ballistic	3.15.2	4.7.9

4.5.3.1 Sampling for inspection of equipment.- Units selected for each group C inspection shall be selected without regard to their quality in accordance with the following.

4.5.3.1.1 Subgroup 1.- For this subgroup, two units from the first 50 production units shall be selected for the first group C inspection. For subsequent group C inspections, two units from each successive 100, or two per month if less than 100 are produced monthly shall be selected.

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4.5.3.1.2 Subgroup II.- For this subgroup, two units from the first 50 production units shall be selected for the first group C inspection. For subsequent group C inspections, two units from each successive 250, or two per month if less than 250 are produced monthly, shall be selected

4.5.3.1.3 Subgroup III.- For this subgroup, two units from the first 50 production units only shall be selected. No subsequent group C inspection for the tests in this subgroup will be required.

4.5.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.5).

4.6 Bounce preconditioning.- Antenna Matching Unit MX-2799( )/VRC shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.7 Service conditions tests.- For tests requiring measurements on Antenna Matching Unit MX-2799( )/VRC in a chamber, dummy loads can be substituted for the antenna whip assembly to facilitate testing.

4.7.1 Temperature test.- The Antenna AT-912( )/VRC shall be subjected to the temperature cycle shown on Standard MIL-STD-169, the following measurements on the antenna matching unit shall be performed at steps 1, 3, 8, and 10. Measurements on the whip antenna assembly during test cycling are not required; however, measurements shall be made on Antenna AT-912( )/VRC external to the test chamber before and after the test cycling.

Test	Paragraph
Cable choke assembly	4.9.1
Voltage standing wave ratio	4.9.2
Feed system efficiency	4.9.3
Power handling	4.9.6

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4.7.2 Moisture-resistance test.- The Antenna AT-912( )/VRC shall be tested as follows:

4.7.2.1 Test conditions.-

(a) The equipment shall be removed from the humidity chamber for measurements which shall be made at ambient temperature and humidity.

(b) The measurements shall be made as rapidly as possible after removal of the equipment from the test chamber.

(c) The seal on Antenna Matching Unit MX-2799( )/VRC shall be broken for all five cycles.

4.7.2.2 Test method.-

(a) Dry at 130°F ±5°F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.7.2.3, and readjust on realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.7.2.3 shall be made during the periods shown on the standard and shall comply with 4.7.2.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F and 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. The equipment shall meet full specified performance for those measurements specified in 4.7.2.3.

4.7.2.3 Performance.- Measurements made during test and specified performance during cycling shall be as follows. The antenna shall meet full specification performance during cycling.

Paragraph	Measurement
Cable choke assembly	4.9.1
Voltage standing wave ratio	4.9.2
Feed system efficiency	4.9.3
Power handling	4.9.6

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4.7.2.4 Failure.- If any equipment fails to meet the performance specified in 4.7.2.3 during cyling, it shall be realined or readjust once. If the equipment then fails to meet 4.7.2.3 or fails subsequently during cycling, it does not pass the test . In addition, if the equipment fails to meet full specification requirements after conditioning, as specified in 4.7.2.2(e), it does not pass the test.

4.7.3 Elevation test.- The Antenna AT-912( )/VRC shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, oprerating and stabilized at room temperature, the voltage standing wave ratio test of 4.9.2 shall be performed on the antenna matching unit.

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above seal level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.7.3(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4±0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.7.3(a). The equipment shall meet specified performance for these measurement.

4.7.4 Immersion test.- The Antenna AT-912( )/VRC shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within 2°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion., the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage.

4.7.5 Orientation test.- The equipment shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left, and right) to an angle of 20 $^{+3}_{-1}$ degrees. During inclination in each plane, the equipment shall meet the VSWR test of 4.9.2.

4.7.6 Sand and dust test.- Antenna AT-912( )/VRC shall be tested as follows:

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4.7.6 Sand and dust test.- (Cont'd)

(a) Place the equipment in the test chamber. Raise and maintain the sand and dust density at 0.1 to 0.5 gram per cubic foot within the test space. The relative humidity shall not exceed 30 percent at any time during the test.

(b) The sand and dust shall be of angular structure and shall have a particle size such that the particles can pass through screens as follows:

Screen, U. S. Standard Sieve Series	Percent
100-mesh	100
140-mesh	98± 2
200-mesh	90± 2
325-mesh	75± 2

(c) Chemical analysis of the sand and dust shall be as follows:

Substance	Percent by weight
SiO_2	97 to 99
Fe_2O_3	0 to 2
Al_2O_3	0 to 1
TiO_2	0 to 2
MgO	0 to 1
Ignition losses	0 to 2

(d) Maintain internal temperature of test chamber at 77° ±4°F for a period of 6 hours, with sand and dust velocity through the test chamber between 100 and 500 feet per minute.

(e) After 6 hours at above conditions, raise the temperature to 160° ±4°F and maintain it at this value for 6 hours. The other conditions shall remain the same.

(f) Remove equipment from test chamber and allow it to cool to room temperature. Accumulated dust shall be removed by brushing, wiping, or shaking, with care taken to avoid introduction of additional dust into the equipment. However, dust shall not be removed by either blast or vacuum cleaning. The equipment shall meet specified performance for the following measurements:

Paragraph	Measurement
4.9.2	VSWR

(g) Open the equipment. There shall be no evidence of deterioration of either external or internal parts which will prevent full operational compliance of the equipment with this specification.

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4.7.7 Vibration.-

4.7.7.1 Test for internal vibration.- Internal vibration of the antenna matching unit measured as follows, to determine conformance to 3.15.1.1:

(a) Secure the antenna matching unit directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this examination the antenna matching unit chassis shall be removed from its case and mounted in a holding fixture duplicating the support provided by the case.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.7.7.2 Vibration, whip assembly (see 3.15.1.2).- The whip assembly shall be vibrated to determine its natural frequency; and then the whip assembly shall be vibrated for two hours at its natural frequency. At the completion of the test, the VSWR test of 4.9.2 shall be performed to determine compliance with 3.15.1.2.

4.7.8 Bounce test.- The Antenna AT-912( )/A[illegible]C shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

(a) Secure the equipment to the 48-day 48-inch vehicular adapter plate, in the same manner as it would be mounted in a vehicle. The adapter plate shall be free of cracks and broken welds, and all unused holes except the 2-inch breather holes shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate.

(b) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

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4.7.8 Bounce test:- (Cont'd)

(c) Place the adapter plate on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to safeguard personnel, provided that the fore-and-aft motion of the adapter plate against the backstop is not restrained.

(d) Operate the package tester, shafts in phase, for a total of 3 hours at 283± 1 rpm. Turn the adapter plate 90 degrees at the end of each 45 minutes, each time in the same direction.

(e) At the end of each 3/4 hour period, the equipment shall be set up with the antenna mounted on a ground plane in an unobstructed area, and the equipment shall be tested for compliance with 3.16.2 and 3.16.6.

4.7.9 Shock test; ballistic.- The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in Specification MIL-S-901. The antenna shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: One each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent, rotation of the equipment under test is permissible.

4.7.10 Shock test; bench-handling.- The chassis-and-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

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4.8 Standard test conditions.- All electrical tests shall be conducted in an electromagnetic environment suitable for making antenna measurements, under the conditions listed below, unless otherwise specified.

Temperature:	Room ambient
Humidity:	Room ambient
Input supply voltage:	25.5 v dc
Input impedance:	50 ohms resistive nominal
Ground plane:	10 ft. x 10 ft. minimum size copper sheet, or equivalent
Frequency:	Accurate to within ± 0.1 mc

Standard test frequencies:

Band	Frequencies (mc)		
1	30.0	31.5	33.0
2	33.0	35.0	37.0
3	37.0	39.5	42.0
4	42.0	44.5	47.5
5	47.5	50.0	53.0
6	53.0	54.5	56.0
7	56.0	58.0	60.0
8	60.0	62.5	65.0
9	65.0	67.5	70.5
10	70.5	73.0	75.95

4.9 Electrical tests.-

4.9.1 Cable choke assembly (see 3.16.1).- With the antenna whip and input cable disconnected, a Wayne Kerr, Type B-801, or equal, Admittance Bridge shall be connected between the spring and the ground (case) of the antenna matching unit, using wires as short as possible. The admittance shall be measured at each standard test frequency by using a Oscillator, General Radio 1211-B, or equal for frequencies up to 50 mc, and an Oscillator, General Radio Type 1215-B, or equal for frequencies above 50 mc, to determine compliance with 3.16.1.

4.9.2 Voltage standing wave ratio test (see 3.16.2).- The test equipment and matching unit shall be connected as shown in Figure 1. The conductance and susceptance shall be measured at each standard test frequency. The VSWR shall be calculated for each frequency, to determine compliance with 3.16.2.

4.9.3 Feed system efficiency (see 3.16.3).- With the test equipment and matching unit connected as shown in Figure 11, 70 watts shall be applied at each test frequency listed below. The Forward Power and Reflected Power shall be measured, and the net Power (P_o) absorbed by the dummy load calculated. The rf voltage (E_o) to ground shall be

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4.9.3 Feed system efficiency (see 3.16.3).-(Cont'd)

measured at the dummy load with the VTVM. The base voltage for an input power of 70 watts shall then be determined by the relationship: $E_{70} = E_o \sqrt{70/P_o}$.

Band	Frequency (mc)		
1	30.0	and	33.0
2	33.0	and	37.0
3	37.0	and	42.0
4	42.0	and	47.5
5	47.5	and	53.0
6	53.0	and	56.0
7	56.0	and	60.0
8	60.0	and	65.0
9	65.0	and	70.5
10	70.5	and	75.95

4.9.4 Input voltage range (see 3.16.4).- The voltage standing wave ratio test of 4.9.2 shall be conducted at frequencies of 30.0 and 75.95 mc with input voltages of 22.0 and 30.0 vdc.

4.9.5 Radiation (see 3.16.5).-The equipment shown in Figure III shall be set up in an unobstructed area with Antenna AT-912( )/VRC whip assembly and a prob antenna (base-fed whip no longer than 6 ft-8 in.) both supported over ground planes at equal heights above ground at a minimum distance from each other of twenty feet. The conductance and susceptance at the rf input terminal of the Antenna Matching Unit shall be measured with the admittance meter to determine the VSWR. An accurately calibrated signal generator tuned to 30.1 mc shall apply rf voltage to the antenna and matching unit under test. Connect the rf detector to the base of the prob antenna and adjust the detector gain to a convenient setting with no overloading and record the signal generator readings. Calculate the corrected power (P_1) from the measured VSWR. The antenna and matching unit shall then be replaced with a standard vertical antenna of proper length to be series resonant at the test frequency. The signal generator voltage shall then be adjusted so that the detector reading shall be the same as that obtained for the antenna and matching unit. Measure the conductance and susceptance at the input terminals of the standard vertical antenna to determine the VSWR. Calculate the corrected power (P_2). Repeat the test at frequencies of 49.8 and 73.3 mc. The ratio of the signal generator output for the standard vertical antenna (P_1) relative to the unit under test (P_2) shall not be less than the values specified in 3.16.5.

4.9.6 Power handling (see 3.16.6).- For the tests of 4.9.6.1 and 4.9.6.2, equipment shown in Figure III shall be used. An rf power of 70 watts shall be applied to the units under test. A reflectometer type device shall be used to measure both the forward and reflected power. Full power shall be applied for 20 minutes at each test frequency. Readings of forward and reflected power shall be made at the start of the test and at 5 minute intervals.

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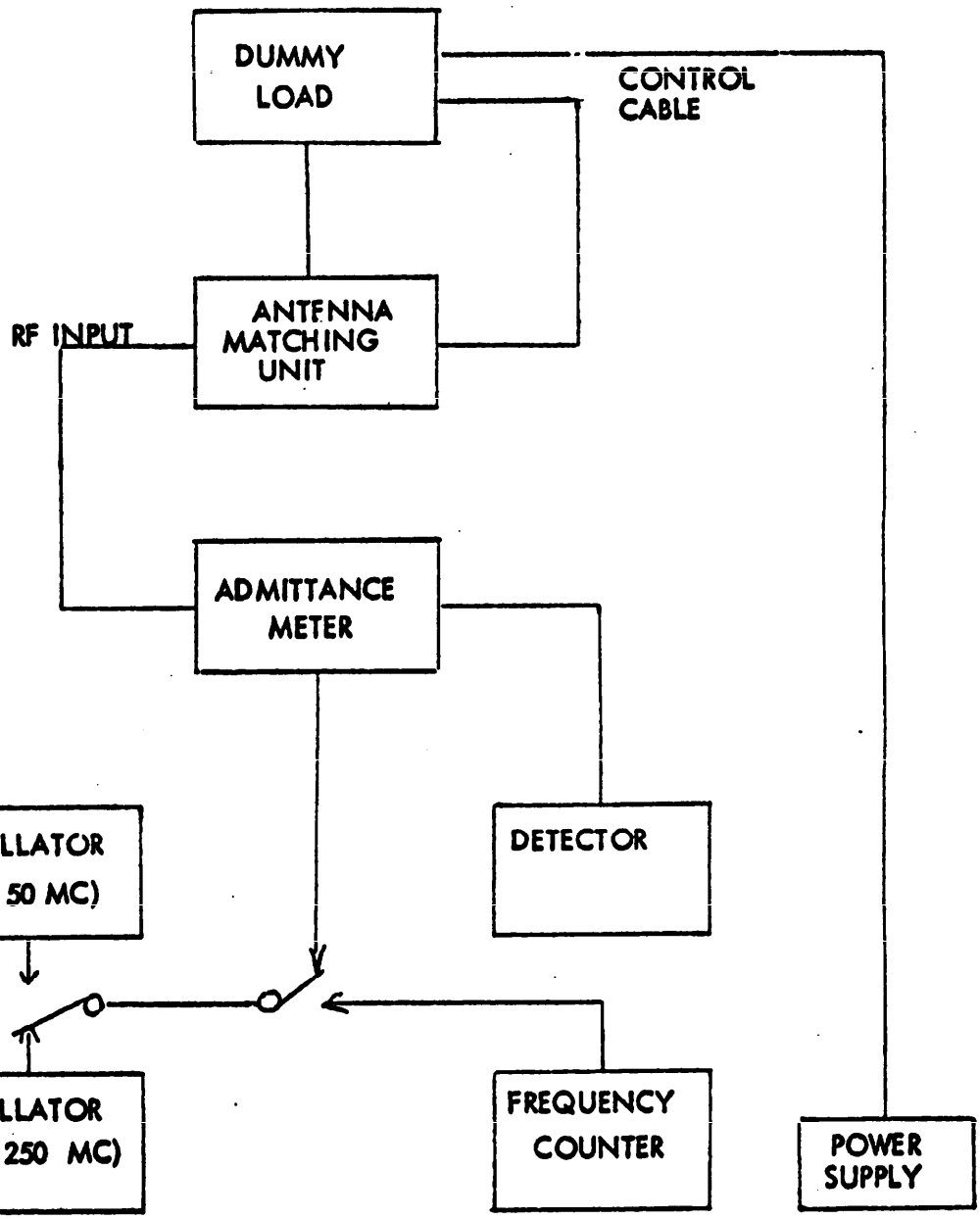
4.9.6.1 Whip antenna and matching unit power test.- The AT-912( )/VRC whip antenna assembly shall be mounted on a ground plane, and the procedure of 4.9.6 shall be performed at test frequencies of 30.1, 49.8, and 73.3 mc to determine compliance with 3.16.6.

4.9.6.2 Matching unit power test.- The Antenna Matching Unit MX-2799( )/VRC shall be terminated with a dummy load to simulate the AT-912( )/VRC whip antenna. The procedure of 4.9.6 shall be performed at test frequencies 31.5, 35.0, 39.5, 44.5, 50.0, 54.5, 58.0, 62.5, 67.5, and 73.0 mc to determine compliance with 3.16.6.

4.10 Impact (see 3.17).- The Antenna AT-912( )/VRC, secured on a vehicle in a normal operating position, shall be subjected to 25 repeated blows in any direction, by a 4 inch x 4 inch oak beam, when the vehicle is travelling at a rate of 25 miles per hour. The point of impact shall be at the midpoint of the whip assembly. The oak beam shall be stationary at the point of impact. After subject to this test, the antenna shall meet the VSWR test of 4.9.2.

4.11 Flexing test (see 3.18).- Base, Antenna Support AB-719( )/VRC complete with rf connector and transmission line, but without Antenna Elements AT-1095( )/VRC and AT-1096( )/VRC, shall be flexed 50,000 cycles, with each cycle consisting of a deflection of the top of the spring from + 45° from the unflexed position and return, to -45° and return. The Base, Antenna Support AB-719( )/VRC shall then be flexed 5000 cycles with one cycle consisting of a deflection from + 90° from the unflexed position and return, to -90° and return. The test shall be performed at temperature of -40°F.

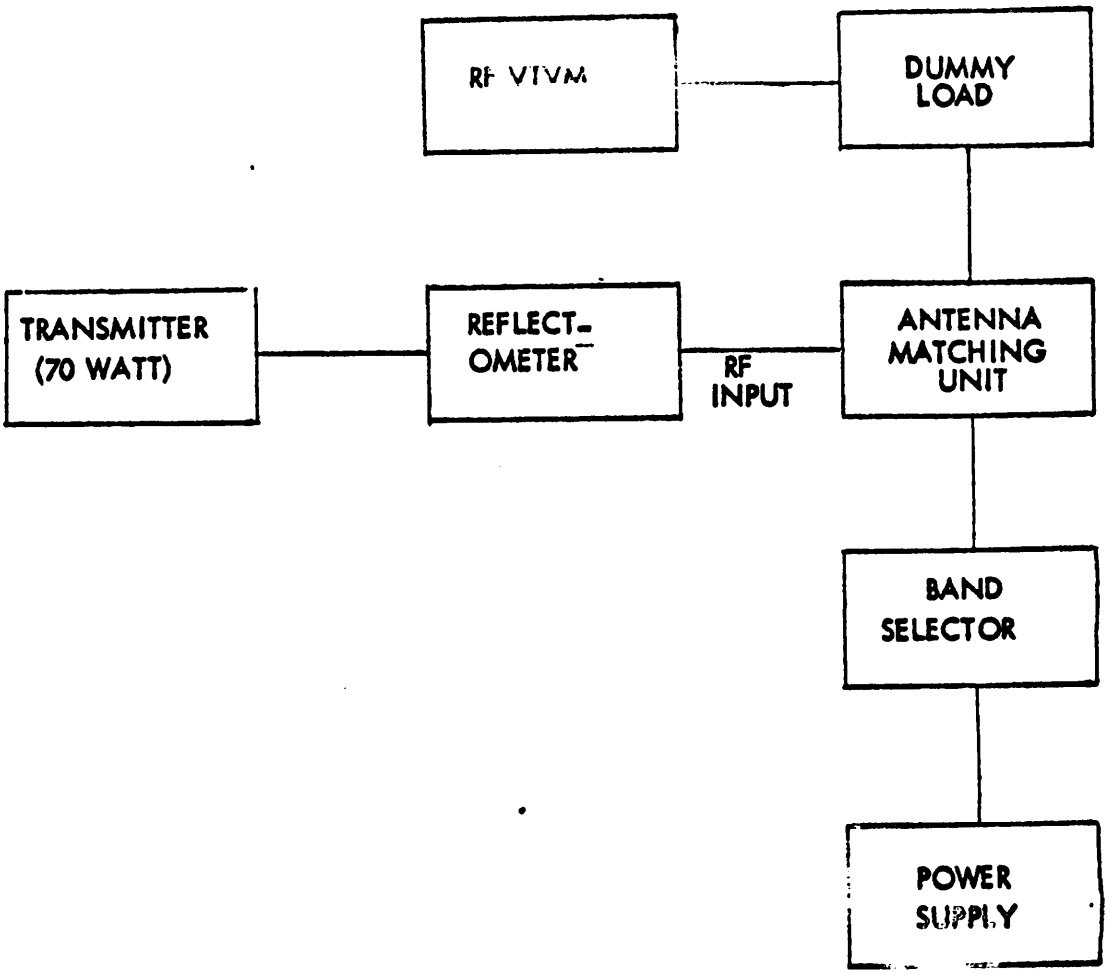
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Admittance meter	Wayne Kerr, B-801, or equal
Detector	General Radio DNT-2, or equal
Frequency counter	Berkley, Nr. 7175, or equal
Oscillator (up to 50 mc)	General Radio, Type 1211-B, or equal
Oscillator (above 50 mc)	General Radio, Type 1215-B, or equal

FIGURE 1

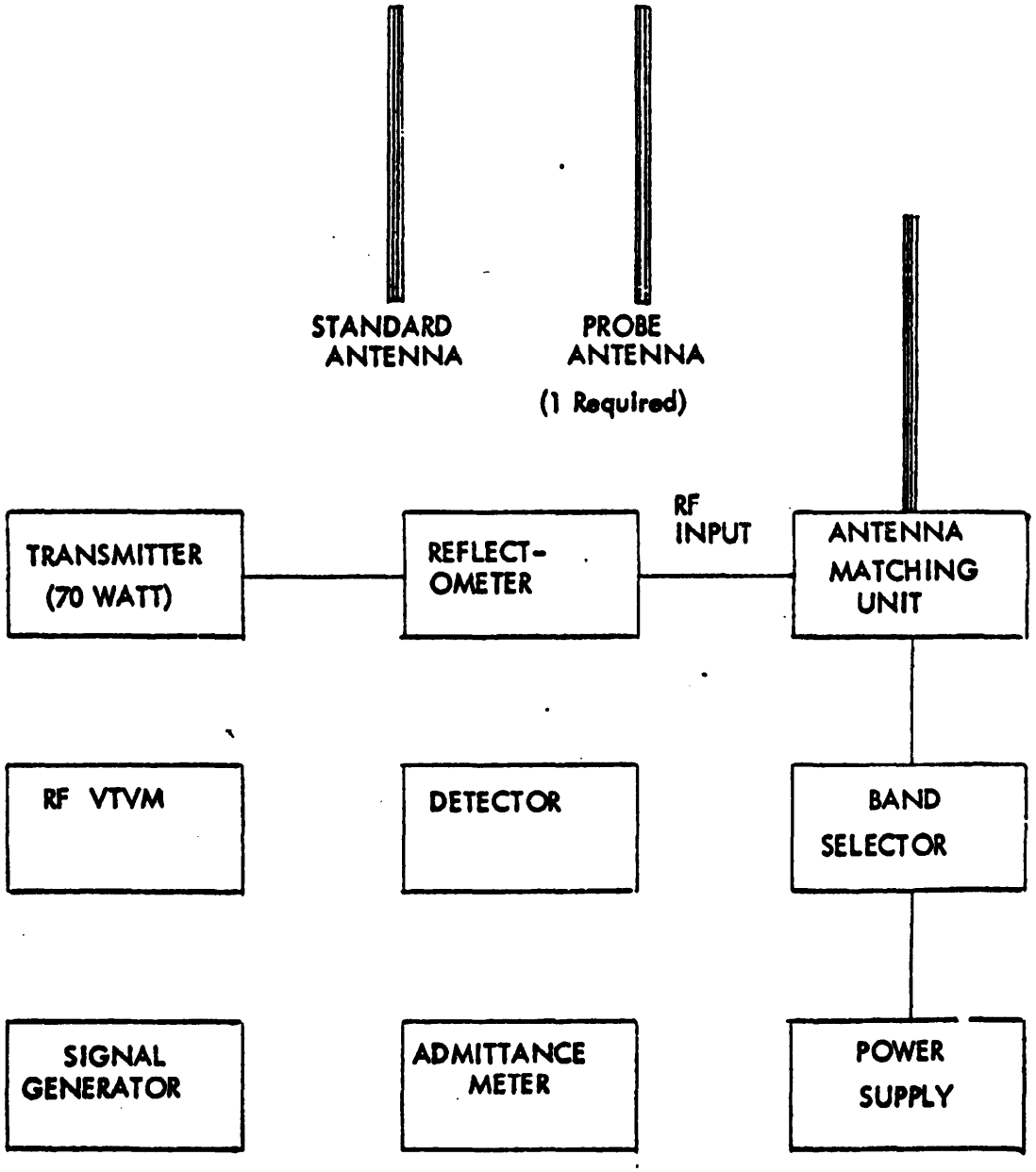
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:F VTVM — General Radio, Type 1800B, or equal
:EFLECTOMETER — Bird, Trueline, Nr. 43, or equal

FIGURE 11

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Reflector	Bird, Trueline, Nr. 43, or equal
RF VTVM	General Radio, Nr. 1800B, or equal
Detector	General Radio, DNT-2, or equal
Signal Generator	Hewlett-Packard, Nr. 608-C, or equal
Admittance Meter	General Radio, 1602-B, or equal

FIGURE III

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4.12 Air-seal test.- The equipment shall be opened and closed again in such a manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment inclosure. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed .01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of .01 pound can be determined readily.

4.13 Visual and mechanical inspection.- The equipment shall be examined for the defects listed in Standard MIL-STD-252 and Drawing SC-A-46694.

4.14 Inspection for interchangeability.- The mechanical dimensions and electrical performance, required by the mechanical and electrical test fixture drawings, shall be measured to determine conformance with the physical and functional interchangeability requirements of 3.21. When a listed dimension or performance requirement is not within specified limits, it shall be considered a major defect. The following dimensions or performance shall be gaged or measured.

(a) Mechanical (dimensions).- The gage points covered by the drawing listed on the applicable gage list listed in (1) below, and the additional gage points specified in (2) below:

(1) (a) Antenna Matching Unit MX-2799( )/VRC: SC-GL-323066

(b) Antenna Element AT-1095( )/VRC: SC-GL-323065

(c) Antenna Element AT-1096( )/VRC: SC-GL-57754

(2) Base, Antenna Support AB-719( )/VRC: The location of holes for four (4) mounting screws.

(b) Electrical performance plug-in impedance matching network assemblies.- Prior to assembly into Antenna Matching Unit MX-2799( )/VRC impedance matching network assemblies shall be tested to determine compliance with the performance requirements specified in the applicable drawings covering construction of the assembly being tested.

4.15 Acceptance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

4.16 Rough handling test (preparation for delivery.- When rough handling test in accordance with MIL-P-116 is required by the contract(see 6.2), the voltage standing wave ratio test (see 4.9.2) shall be conducted to determine freedom from operational malfunction caused by rough handling.

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5. PREPARATION FOR DELIVERY

5.1 -Preservation and packaging.-

5.1.1 Level A.- Antenna AT-912( )/VRC shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and as described herein.

5.1.1.1 Unit packaging.- Antenna AT-912( )/VRC shall be unit packaged Method IC-5 as follows: Cushion each antenna on all surfaces with cells or pads or both, fabricated of double-faced corrugated fiberboard and designed to absorb the shock of impact encountered in handling and transit. Fiberboard shall have water-resistant qualities equal to that of the fiber box in which the cusioned items will be placed. Place the cushioned items within a close-fitting regular slotted style, class 2, W5c fiberboard box conforming to Specification PPP-B-636. Box closure shall be in accordance with the referenced box specification.

5.1.2 Level C.- Antenna AT-912( )/VRC shall be preserved andpackaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.- A quantity of Antenna AT-912( )/VRC packaged as specified in 5.1.1 shall be packed for shipment within a class 2, style 4 nailed wood, conforming to Specification PPP-B-621. The gross weight of any shipping container shall not exceed 200 pounds. Fabricate the box to fit the contents snugly. Box closure shall be in accordance with the box specification.

5.2.1.1 Metal strapping.- Shipping containers shall be strapped in conformance with the requirements of the appendix of the reference container specification only for direct shipment to ports.

5.2.2 Level B.- Antenna AT-912( )/VRC shall be packed for shipment as specified in 5.2.1 except that the nailed wood box shall be class 1, style 2.

5.2.3 Level C.- The pack shall be as specified in 5.2.2.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Military Standard MIL-STD-129.

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5.4 Pilot pack.- When level A packaging and packing is specified above, one acceptable model of Antenna AT-912( )/VRC will be pilot packed in a manner that will pass the preproduction tests prescribed in Specification MIL-P-116. Illustration of the packaging and packing procedure, together with Bills of Materials, will be prepared and furnished in accordance with the provisions of Drawing SC-A-362100.

6. NOTES

6.1 Intended use.- Antenna AT-912( )/VRC is intended for use with Radio Sets AN/VRC-12, AN/VRC-43 through AN/VRC-49, AN/VRC-53, and AN/GRC-125.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(f) Marking and shipping of samples.

(g) Place of final inspection

(h) Technical literature required. (See 3.22).

(i) Quantity of running spare parts required. (See 3.22).

6.3 Color.- The color chip furnished by the procuring agency (3.8) will be color chip No. X-24087, and may be obtained upon request to Commanding General, U. S. Army Electronics Materiel Agency, 225 South Eighteenth Street, Philadelphia 3, Pennsylvania, ATTN: SELMA-J4b.

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6.4 Definition of level B.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.5 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2.)

6.6 Location of air-seal test.- It is desirable that the air-seal test (4.11) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.7 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AT-912W/VRC, The contractor should apply for nomenclature in accordance with the applicable clause in the contract.(See 1.1.)

6.8 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A-- The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B-- That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

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6.8 Verification Inspection.- (Cont'd)

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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